

JACOB ARAGAO

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WORK EXPERIENCE

NationGraph

Software Engineer Intern | TypeScript, React, Python, Go, PostgreSQL, Tailwind

05/2026 – 08/2026

- Architected an end-to-end **AI recommendations engine** with Go, Redis, and WebSockets; orchestrated 3 parallel **streamed LLM calls** to slash round-trip latency by ~50%.
- Spearheaded the **Signal Inbox** full-stack redesign to eliminate 504 timeouts, dropping bulk-triage latency from ~10s to **sub-second** and query runtimes to ~30ms via PostgreSQL index tuning.
- Drove product direction for **Compass** (agentic AI chat); pitched a UX roadmap to executive leadership to decouple complex workflows into dedicated dashboards while shipping high-intent home page prompts to drive daily engagement.
- Hardened application security across **13 endpoints**, eliminating **IDOR vulnerabilities** on critical surfaces by refactoring API routes to derive identity from verified **JWT claims**.
- Accelerated frontend velocity by authoring **16 design-system components** (~26% of library) in React/TypeScript and migrating the flagship application to **Tailwind CSS v4**.

CareHive Health

Software Engineering Intern | TypeScript, SolidJS, Node.js, PostgreSQL, Vitest

05/2025 – 08/2025

- Built and shipped a full-stack **Reminder Engine** to re-engage members; scaled architecture from **0 to 9 services** with per-service checkpoints and deep links.
- Designed a custom **sorting algorithm** to rank and surface the highest-value user actions based on dynamic priority metrics.
- Refactored core business logic into isolated components, scaling unit test suites in **Vitest** to maintain **97% test coverage**.

Clearer.io

Junior Software Engineer | React, Node.js, PostgreSQL, Redis, Stripe

09/2024 – 04/2025

- Built data ingestion workflows using **Node.js** and **webhooks** to sync real-time tracking data from ShipStation, increasing shipment visibility by **50%**.
- Independently designed and deployed Instant Exchanges, integrating **Stripe** payments directly into the return flow; streamlined online exchanges for **39%** of users.

Software Engineering Intern | Node.js, AWS S3, Sequelize ORM

05/2024 – 08/2024

- Identified and patched a critical **live checkout vulnerability** by enforcing strict server-side price and cart validation, preventing client-side price tampering and direct financial loss.
- Orchestrated zero-downtime **PostgreSQL** schema migrations via **Sequelize ORM**, remediating production sync failures for over **900 active users**.

EDUCATION

McGill University, Montreal, Quebec

08/2022 – 04/2027 (Expected)

Bachelor of Engineering in Software Engineering (Co-op)

- **Academic Distinction:** Placed in the top 15% of the Faculty of Engineering (2024–2025).

PROJECTS

FantasyGuide

youtu.be/EM8tWZOGKWI

- Engineered a GenAI data pipeline in **Python** that ingests raw player statistics via **BeautifulSoup**, normalizes records using **Pandas**, and constructs structured prompt payloads for the **OpenAI API**.
- Architected a full-stack interface with **React** and **Flask**, streaming dynamic player projections and real-time AI draft recommendations to active users.

Vantage

youtu.be/O69Y7m8nO_I

- Architected a real-time multimodal voice assistant using **Django Channels** and **Gemini 2.5 Flash API**, implementing full-duplex **WebSockets** for simultaneous audio capture and screen context injection.
- Containerized the application with **Docker Compose**, orchestrating the Daphne ASGI server and MongoDB.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript, Python, Go, SQL, HTML, CSS, Java, C
- **AI & Data:** LLM APIs, Prompt Engineering, Scikit-learn, Pandas, NumPy
- **Full Stack:** React, SolidJS, Tailwind CSS, Node.js, Express, Django (Channels), Flask, WebSockets
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, AWS (S3, CloudWatch), Docker
- **DevOps & Testing:** Git, GitHub, GitLab CI/CD, Jira, Grafana, Vitest, Jest, pytest-asyncio